

Note 8.2 Trigonometric Integrals

Products of Powers of Sines and Cosines

- **Strategy:**

For the integrals of the form:

$$\int \sin^m x \cos^n x dx$$

where m and n are nonnegative integers, then we can divide into cases based on m and n being odd or even.

- If one of m and n is odd, then we can combine the single $\sin x$ or $\cos x$ with dx in the integral. Then we can use [the substitution](#) to solve.
- If both m and n are even, then we need to substitute

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \cos^2 x = \frac{1 + \cos 2x}{2}$$

to reduce the integrand to one in lower powers of $\cos 2x$.

- **Example:**

- Evaluate $\int \sin^3 x \cos^2 x dx$.

This is in the first case.

$$\begin{aligned} \int \sin^3 x \cos^2 x dx &= \int \sin^2 x \cos^2 x \sin x dx \\ &= \int (1 - \cos^2 x) \cos^2 x (-d(\cos x)) \\ &= - \int (1 - u^2) u^2 du \\ &= \int (u^4 - u^2) du \\ &= \frac{u^5}{5} - \frac{u^3}{3} + C = \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C \end{aligned}$$

- Evaluate $\int \cos^5 x dx$.

This is in the first case.

$$\begin{aligned}
 \int \cos^5 x \, dx &= \int \cos^4 x \cos x \, dx \\
 &= \int (1 - \sin^2 x)^2 d(\sin x) \\
 &= \int (1 - u^2)^2 du \\
 &= \int (u^4 - 2u^2 + 1) du \\
 &= \frac{\sin^5 x}{5} - \frac{2 \sin^3 x}{3} + \sin x + C
 \end{aligned}$$

- Evaluate $\int \sin^2 x \cos^4 x \, dx$.

This is in the second case.

$$\begin{aligned}
 \int \sin^2 x \cos^4 x \, dx &= \int \frac{1 - \cos 2x}{2} \left(\frac{1 + \cos 2x}{2} \right)^2 dx \\
 &= \frac{1}{8} \int (1 - \cos 2x)(1 + 2 \cos 2x + \cos^2 2x) dx \\
 &= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) dx \\
 &= \frac{1}{8} \left[x + \frac{1}{2} \sin 2x - \int (\cos^2 2x + \cos^3 2x) dx \right]
 \end{aligned}$$

then by the [formula of cosine's n power](#),

$$\int \sin^2 x \cos^4 x \, dx = \frac{1}{16} \left(x - \frac{1}{4} \sin 4x + \frac{1}{3} \sin^3 2x \right) + C$$

Integrals of Powers of $\tan x$ and $\sec x$

- **Idea:**

$\tan x$ and $\sec x$ have the relationship:

- $\tan^2 x = \sec^2 x - 1$, $\sec^2 x = \tan^2 x + 1$.
- $\frac{d}{dx} \tan x = \sec^2 x$, $\frac{d}{dx} \sec x = \sec x \tan x$.

Products of Sines and Cosines

- **Strategy:**

For the integrals $\int \sin mx \sin nx \, dx$, $\int \sin mx \cos nx \, dx$, $\int \cos mx \cos nx \, dx$, we can use the

formulas to transform the products into the sum:

$$\sin mx \sin nx = \frac{1}{2} [\cos(m-n)x - \cos(m+n)x]$$

$$\sin mx \cos nx = \frac{1}{2} [\sin(m-n)x + \sin(m+n)x]$$

$$\cos mx \cos nx = \frac{1}{2} [\cos(m-n)x + \cos(m+n)x]$$

Exercises

- P466 21, 41, 47